

DECODELABS • PROJECT 2 EDA

E-COMMERCE EXPLORATORY DATA ANALYSIS

*Revenue drivers, channel performance, order risk, and anomalies across
DecodeLabs' order history*

1,200 ORDERS • JAN 2023 – JUN 2025 • 14 COLUMNS

Problem Statement & Methodology

What patterns, trends, and outliers exist in DecodeLabs' order data? This EDA identifies revenue drivers, order-status distribution, channel performance, and anomalies to guide business decisions.

- Descriptive statistics — mean, median, std dev, quartiles for all numeric columns
- Monthly & annual revenue trend analysis
- Segmentation by product, status, payment, and referral source
- IQR outlier detection on TotalPrice
- Pearson correlation across Quantity, UnitPrice, TotalPrice, ItemsInCart

TOTAL REVENUE

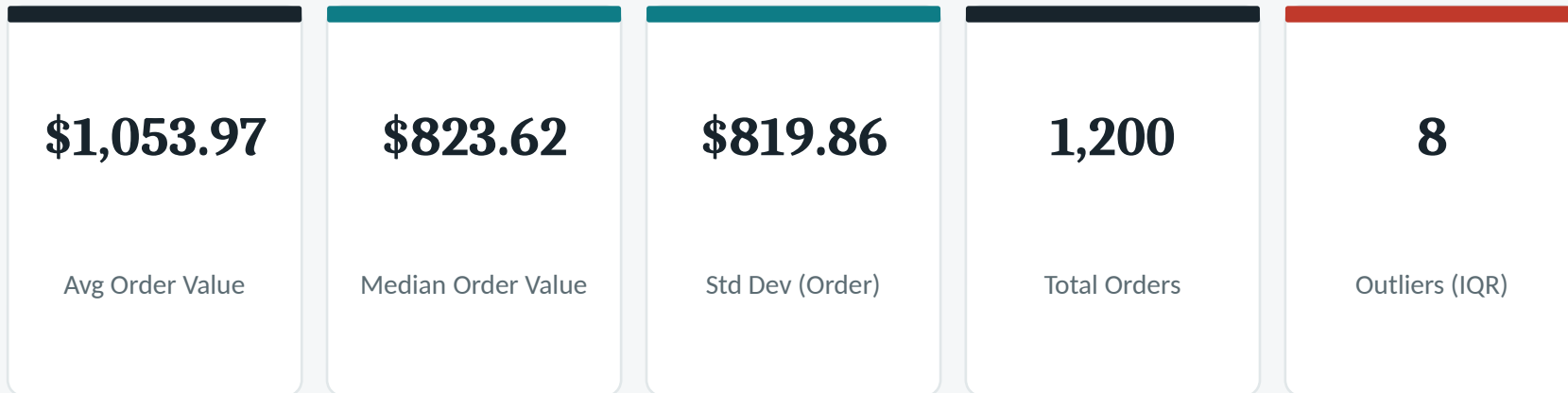
\$1.26M

\$1,264,761.96 exact

Across 1,200 Orders
Jan 2023 – Jun 2025

*Avg order value \$1,053.97 — but
trending down year over year*

Order-Level Summary Metrics

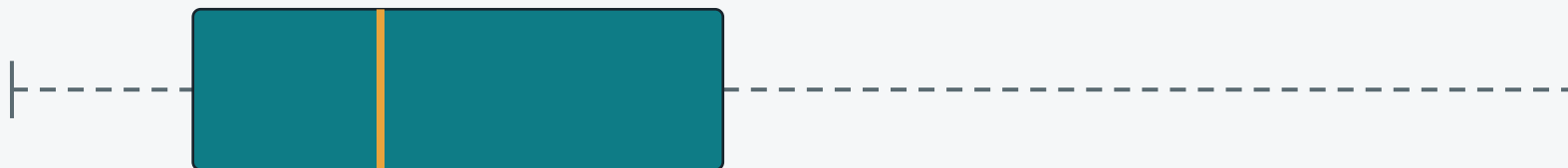


Order value is right-skewed — the mean (\$1,053.97) sits well above the median (\$823.62), pulled upward by high-value VIP orders.

DESCRIPTIVE STATISTICS

Order Value Distribution (TotalPrice)

Five-number summary across all 1,200 orders



MIN

\$11.39

Q1

\$410.52

MEDIAN

\$823.62

Q3

\$1,578.48

MAX

\$3,456.40

Mean

\$1,053.97

Std Dev

\$819.86

IQR (Q3-Q1)

\$1,167.96

KEY INSIGHT

What Actually Drives Order Value?

Pearson correlation (r) across Quantity, UnitPrice, TotalPrice, and ItemsInCart

STRONGEST DRIVER

$$r = 0.717$$

UnitPrice → TotalPrice

Price per unit is the single strongest predictor of what an order is worth

SECOND DRIVER

$$r = 0.615$$

Quantity → TotalPrice

Quantity ordered matters, but less than the price of the item itself

NOT CORRELATED

$$r = 0.001$$

ItemsInCart ↔ UnitPrice

Browsing/cart-building behaviour is independent of how much customers spend

SEGMENTATION

Revenue by Product

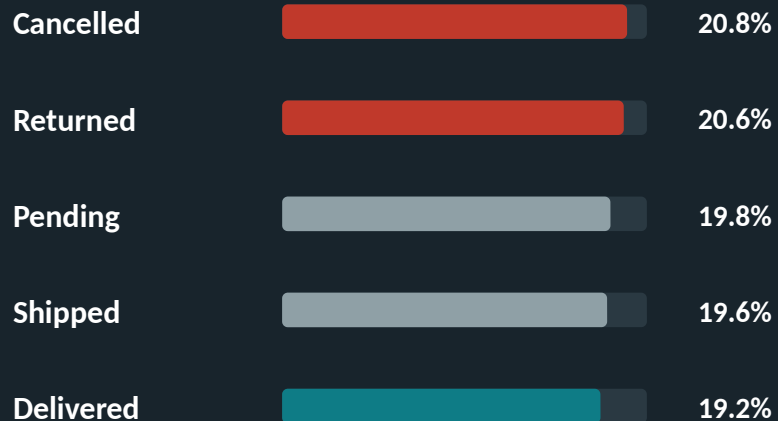
Chair and Printer are effectively tied as the top revenue generators; Phone is lowest



KEY INSIGHT

41.4% of Revenue Is at Risk

ORDER STATUS BREAKDOWN



41.4%

Cancelled + Returned

Only 19.2% of orders are ultimately Delivered — the smallest of all five statuses. Nearly as much revenue is lost to cancellations and returns as is successfully fulfilled.

SEGMENTATION

Revenue by Referral Source



BEST CHANNEL

Instagram

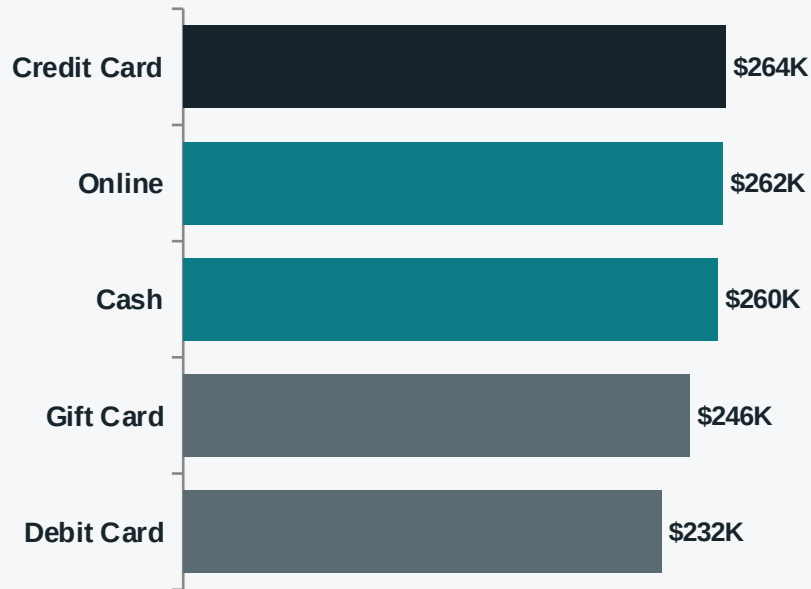
259 orders • \$275K revenue

Drives both the most orders and the highest total revenue of any channel — Referral is the weakest performer by a wide margin.

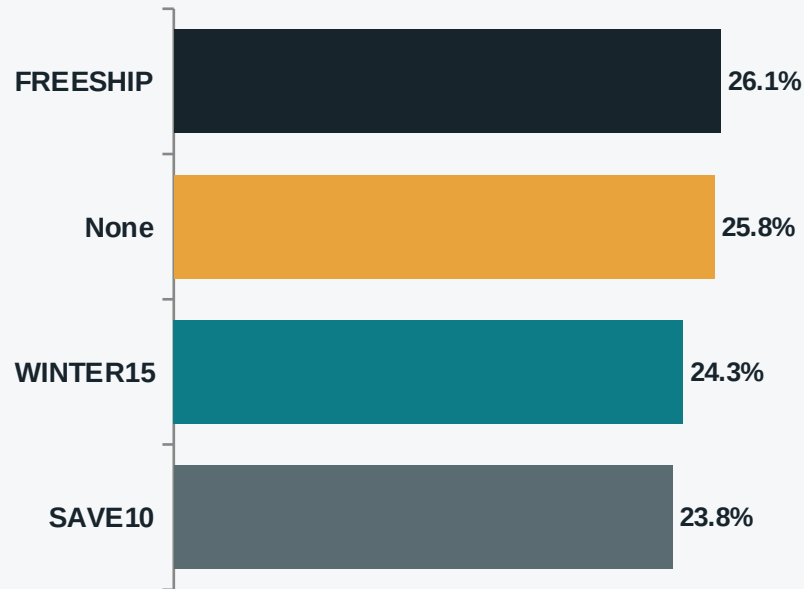
SEGMENTATION

Payment Methods & Coupon Usage

REVENUE BY PAYMENT METHOD (\$K)



COUPON USAGE (% OF ORDERS)

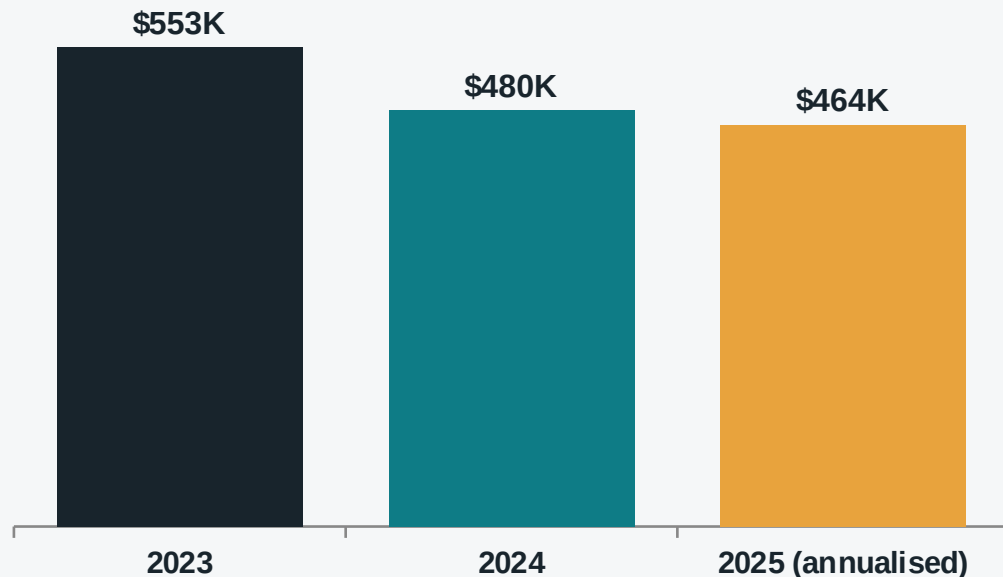


Credit Card leads on revenue; Online leads on order count (258). FREESHIP is the most-used coupon; a quarter of orders use no coupon at all.

TREND ANALYSIS

Revenue Is Declining Year Over Year

2025 figures are annualised from 6 months of data (Jan-Jun)



AOV IS SLIPPING TOO

\$1,084 → **\$1,004**

2023

2025

Average order value has fallen every year — both fewer orders and smaller baskets are dragging revenue down.

TREND ANALYSIS

Monthly Revenue, Jan 2023 – Jun 2025

Volatile month-to-month, but the underlying level has drifted lower across all 30 months



OUTLIER DETECTION

The 8 Outliers Are a VIP Segment, Not Errors

IQR METHOD

Q1 (25th Pct)	\$410.52
Q3 (75th Pct)	\$1,578.48
IQR (Q3-Q1)	\$1,167.96
Upper Fence	\$3,330.41

8

orders flagged above the upper fence
of \$3,330.41

VERDICT

All 8 outliers share the same profile: Quantity = 5, UnitPrice \$666-\$692.

These are legitimate high-volume orders by VIP customers — not data-entry errors. Order statuses span Delivered, Shipped, Returned, and Cancelled, spread across five different products.

RECOMMENDATION

Flag as a “VIP Segment” for targeted retention campaigns.

RECOMMENDATIONS

Where to Focus Next

01

Investigate the 41.4% Cancellation + Return Rate

Identify top cancellation reasons by product to stem the largest source of revenue leakage.

02

Prioritise Instagram Marketing Spend

It generates the highest order volume and revenue of any channel — the best ROI across all channels tested.

03

Introduce Loyalty Incentives for VIP (Qty=5) Customers

The outlier segment represents high-value, high-volume buyers worth a dedicated retention programme.

04

Investigate the Declining Average Order Value

AOV has fallen every year — from \$1,084 (2023) to \$1,004 (2025) — and warrants root-cause analysis.

05

Run Targeted SAVE10 Campaigns

SAVE10 underperforms FREESHIP despite similar usage rates — a targeted push could close the gap.

CONCLUSION & CALL TO ACTION

- \$1.26M in revenue across 1,200 orders, but 41.4% is tied up in cancellations and returns
- UnitPrice is the strongest driver of order value; Instagram is the strongest acquisition channel
- Revenue and average order value are both declining year over year and need root-cause investigation
- The 8 statistical outliers are a legitimate VIP segment, not data errors

IMMEDIATE PRIORITY

Cut the Cancellation & Return Rate

41.4% of orders never convert to delivered revenue

BUILD ON STRENGTH

Double Down on Instagram & VIPs

Best channel and highest-value customer segment identified

Fixing order-completion risk while doubling down on Instagram and VIP retention offers the clearest path back to revenue growth.